Data Analytics Report

Week 2 – Cohort Analysis Report and Visual Insights

AI Powered Data Analysis Remote Internship

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Team name: Team 6

Dataset: Student Participation Dataset

Contributions

Genevieve Ah Shene – Task Overview and Opportunity Category Analysis

Busayo Salau, Chiamaka Joy & Rejoice Daniel - Data cleaning, Feature Engineering and Documentation.

Vritaant Bajaj & Yonnie Ntsubane & Madunagu Adaeze - Exploratory Data Analysis and Visualizations

Chiamaka Joy – Forecasting Signals

Usman Bin Yousuf – Quarterly Analysis on Applications & Compilation of the final report.

Overview

For this week's deliverable, the focus is on Cohort Analysis and Visual Insights, using Exploratory Data Analysis (EDA) to identify patterns, trends, and relationships within the dataset. The analysis involves key questions such as which opportunity categories have the highest participation, how participation changes over time, whether there is seasonal patterns, and how learners' participation is distributed across different statuses.

The analysis also identifies forecasting signals, including variables that are likely to influence future predictions and strong indicators within learner participation. Key observations are highlighted to support potential future modeling and provide a foundation for identifying trends that may continue over time.

As part of the analysis, feature engineering was also conducted to prepare the dataset for future modeling. This included creating new variables from existing data and extracting quarter information from date-related variables to support analysis of participation patterns across different periods. These engineered features help provide additional structure to the dataset and make it easier to identify relationships, trends and seasonal behaviour.

Visualizations are used to support and communicate the results clearly, helping provide a better understanding of learner participation, category performance, and important trends within the dataset.

Cleaning Steps Taken This Week:

1) Some rows that could not be parsed as datetime in Learner signup datetime and apply datetime were removed (about 57 rows).

2) All Remaining rows in learner signup datetime were standardized to a single format.

3) opportunity start datetime had about 3000+ rows that were empty and since the column was needed to create new features, the empty values were replaced with apply datetime +30. The reason for this is because a learner is likely to start a program on an average of 30 days after application.

4) All these were done using a combination of worksheet formulas and power query.

FEATURE ENGINEERING

Thirty new features were created from the existing variables which are given in the table below

FEATURES	DESCRIPTION	ANALYTICAL RELEVANCE
Learner SignUp Date	The date the learner registered on the platform.	To detect timing patterns.
SignUp Year	The year the learner registered on the platform.	To spot yearly sign-up trends.
SignUp Month	The month the learner registered on the platform.	To spot seasonal sign-up trends
SignUp Day	The day the learner registered on the platform.	To spot day of the month sign up patterns.
Learner SignUp Time	The time the learner registered on the platform.	To identify peak sign-up hours.
Apply Date	The date the learner applied for the program.	To detect timing patterns in applications.
Apply Year	The year the learner applied for the program.	To spot yearly application trends.
Apply Month	The month the learner applied for the program.	To spot seasonal application trends.
Apply Day	The day the learner applied for the program.	To spot day of month application patterns.

Apply Time	The time the learner applied for the program.	To identify peak application hours.
Days of Application after Signup	The number of days from signup to apply date.	To measure how quickly learners apply after signing up.
Opportunity End Date	Closing date of the opportunity.	To detect timing patterns in program closures.
Opportunity End Year	Closing year of the opportunity.	To spot yearly closure trends.
Opportunity End Month	Closing month of the opportunity.	To spot seasonal closure trends.
Opportunity End Day	Closing day of the opportunity.	To check for day-of-month closure patterns.
Opportunity End Time	Closing time of the opportunity.	To identify peak closure hours.
Opportunity Start Date	The date the opportunity started.	To detect timing patterns in program launches.
Opportunity Start Year	The year the opportunity started.	To spot yearly launch trends.
Opportunity Start Month	The month the opportunity started.	To spot seasonal launch trends.
Opportunity Start Day	The day the opportunity started.	To check for day-of-month launch patterns.
Opportunity Start Time	The time the opportunity started.	To identify peak launch hours.
Duration of Program (Days)	The length of opportunity in days.	To compare program lengths and their effect on outcomes.
Duration of Program (Month)	The length of opportunity in month.	To compare program lengths on a broader scale.
Days between opportunity start date and apply date	The number of days it took from application to the program commencement.	To measure how far in advance learners apply before a program starts.
Months between opportunity start date and apply date	The number of months it took from application to the program commencement.	To measure application lead time on a broader scale.
Age of Learner	Age of the learner.	To analyze participation patterns across age groups.
Quarter Application Date	Period of time, every three months in which learners applied for the program.	To spot quarterly trends in application.
Quarter Learners Sign up Date	A three-month period of time through the year learners registered for the program.	To spot quarterly trends in learners sign up.
Quarter Opportunity Start Date	A three-month period of time through the year in which learners started the program.	To spot quarterly trends in program launches.
Quarter Opportunity	A three-month period of time	To spot quarterly trends in

End Date	through the year in which learners ended the program.	program completion.

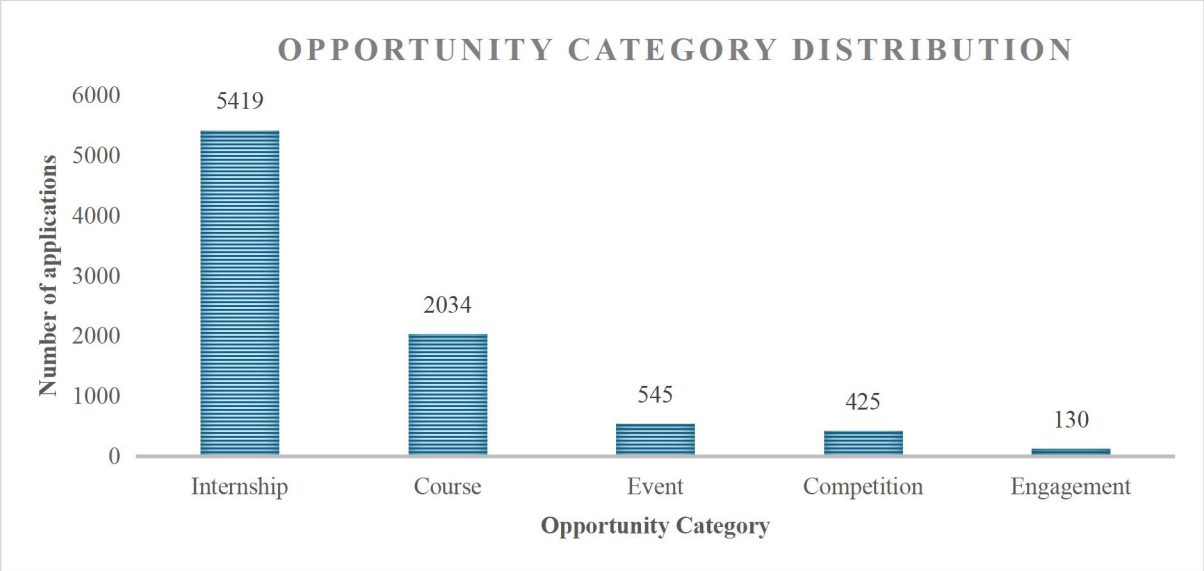
Exploratory Data Analysis

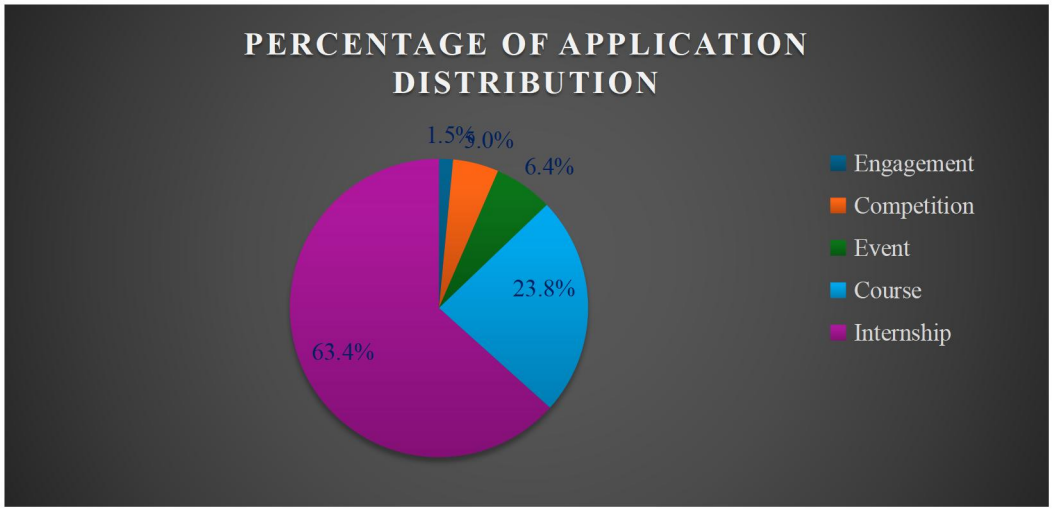
Summarizing Participation

Opportunity Breakdown and Total Application Distribution

The opportunity breakdown provides an overview of the distribution of applications across the various opportunities recorded in the dataset.

The total application distribution represents the number of applications associated with each opportunity category. To obtain this, the opportunity variables are placed in a PivotTable, while the application record is summarized using count.





Distribution of the 8553 application records across opportunity categories and its percentage record in the clean dataset

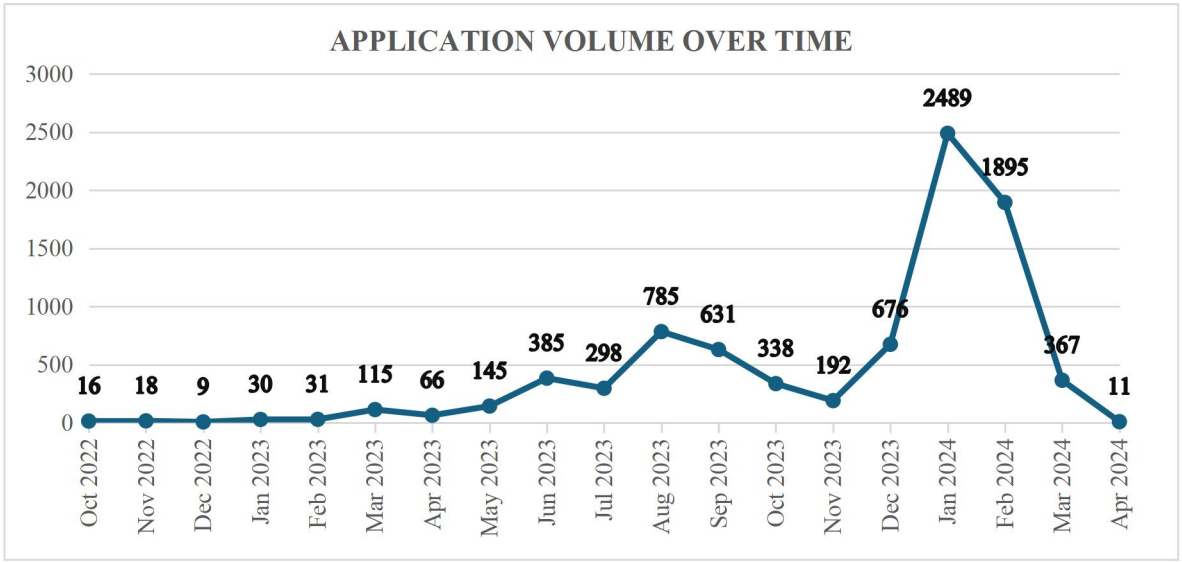
Interpretation:

The charts provide a visual comparison of the total number of applications received by each Opportunity. It shows that applications were heavily concentrated on Internship opportunity with 63.4% and course opportunity with 23.8% of the total application records, event 6.4% and competition 5.0% have fewer applications with engagement 1.5% being the lowest application and the least interested opportunity.

This indicates that Internship is the most dominant opportunity in the application records.

Application Over Time

This examines how the number of applications in the dataset changes across different periods and identify when application activity was low, high, increasing, decreasing or fluctuating. It's achieved by grouping the Apply Date/Time to Month/Year.



Note: Each point represents the number of application records whose apply dates fall within the month.

Interpretation:

The analysis of Apply Month/Year shows significant changes in application volume over time. Application activity was relatively low between October 2022 and February 2023 but increased considerably from March 2023 onwards. A fluctuation occurred between September 2023 and November 2023 before increasing again in December, which indicate that application activity wasn't steadily increasing every month. There was a high volume of application in January 2024 followed by decline in February 2024 and a high reduction from March through to April.

Consistency Trends

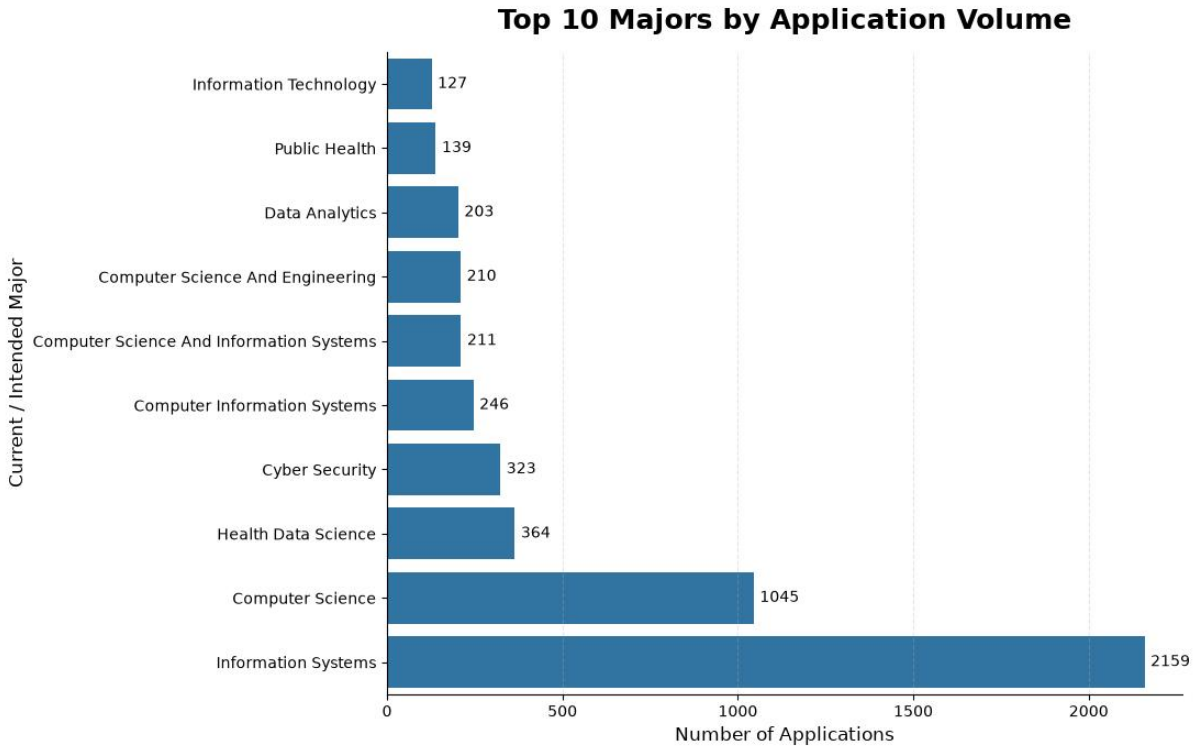
Top 10 Majors by Application Volume

This visualization shows the 10 most common current or intended majors among the applicants in the dataset. Each bar represents the number of applications associated with a particular major, allowing us to compare participation across different academic backgrounds.

From the visualization, Information Systems has the highest number of applications (2,159), followed by Computer Science (1,045). The remaining majors have considerably lower application volumes, with Health Data Science and Cyber Security being the next more represented fields.

This helps us understand which academic backgrounds have the strongest representation in the applicant population and whether participation is concentrated in a few specific fields.

One important point is that similar majors, such as Computer Science, Computer Science and Engineering, and Computer Science and Information Systems, are treated as separate categories because they appear separately in the dataset.





Monthly Application Consistency Trends

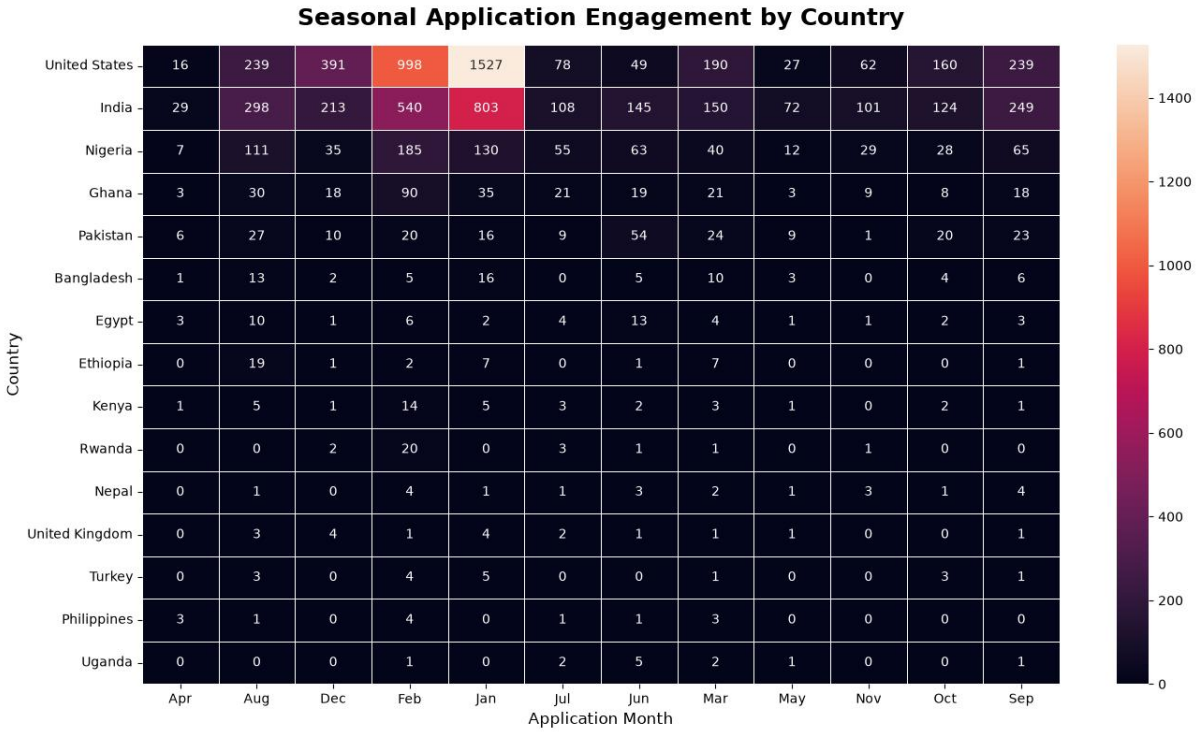
This visualization represents the monthly application volume recorded in the dataset and is used to understand how consistently users applied over time. The line chart makes it easier to observe changes in application activity from one month to another and identify periods of higher or lower engagement.

In the earlier months, the number of applications remains relatively low, with gradual changes in activity. During 2023, application volume begins to increase, although the pattern

continues to fluctuate rather than following a steady upward trend. This indicates that participation varied across different months.

The most noticeable change occurs around the beginning of 2024, where application activity rises sharply and reaches its highest point in January 2024, with approximately 2,489 applications. This is followed by a decline in February and a much sharper reduction in the following months.

Overall, the visualization indicates that application activity was not consistent throughout the entire period. There were significant fluctuations, including a major spike in early 2024. The graph therefore helps identify periods of increased engagement and periods of relatively low activity, which can be useful for investigating possible seasonal patterns or changes in application behaviour.



Seasonal Application Engagement by Country

This visualization shows how application activity varies across different months for the top countries represented in the dataset. The heat map uses colour intensity to highlight months with higher or lower numbers of applications, making it easier to identify patterns in engagement across countries.

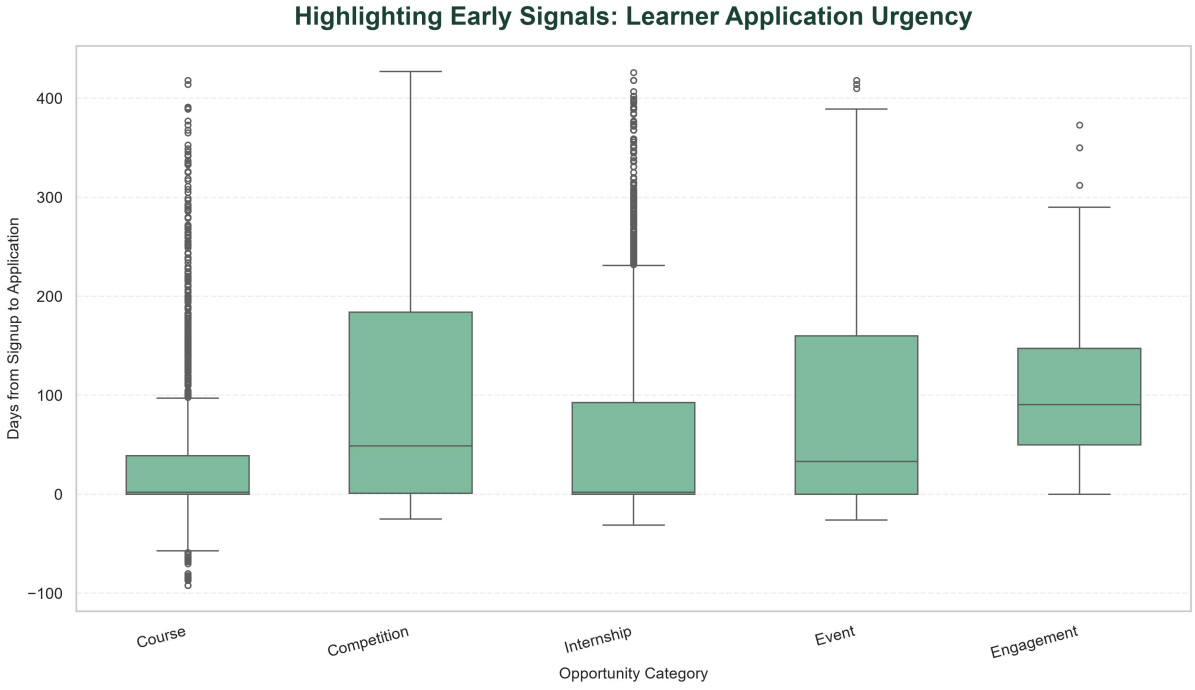
The United States and India have the highest overall application volumes. For the United States, application activity is particularly high in January (1,527) and February (998), while India also shows its highest activity in January (803) and February (540). Other countries have considerably lower application volumes, with their activity distributed across different months.

Overall, the visualization indicates that application engagement varies considerably by country and month, with January and February showing particularly high activity among the two largest applicant groups. This suggests that engagement is not evenly distributed throughout the year and that certain periods have substantially higher application activity.

Highlighting Early Insights

Learner Application Urgency

This visualization analyses the conversion speed of platform users by measuring the exact number of days elapsed between initial user sign-up and final application submission. To evaluate behavioural differences across offerings, the dataset was segmented by opportunity categories, including Course, Competition, Internship, Event, and Engagement. The distribution spread, median lead times and extreme behaviours were mapped using a box plot structure to highlight conversion momentum.



Key Insights & Findings

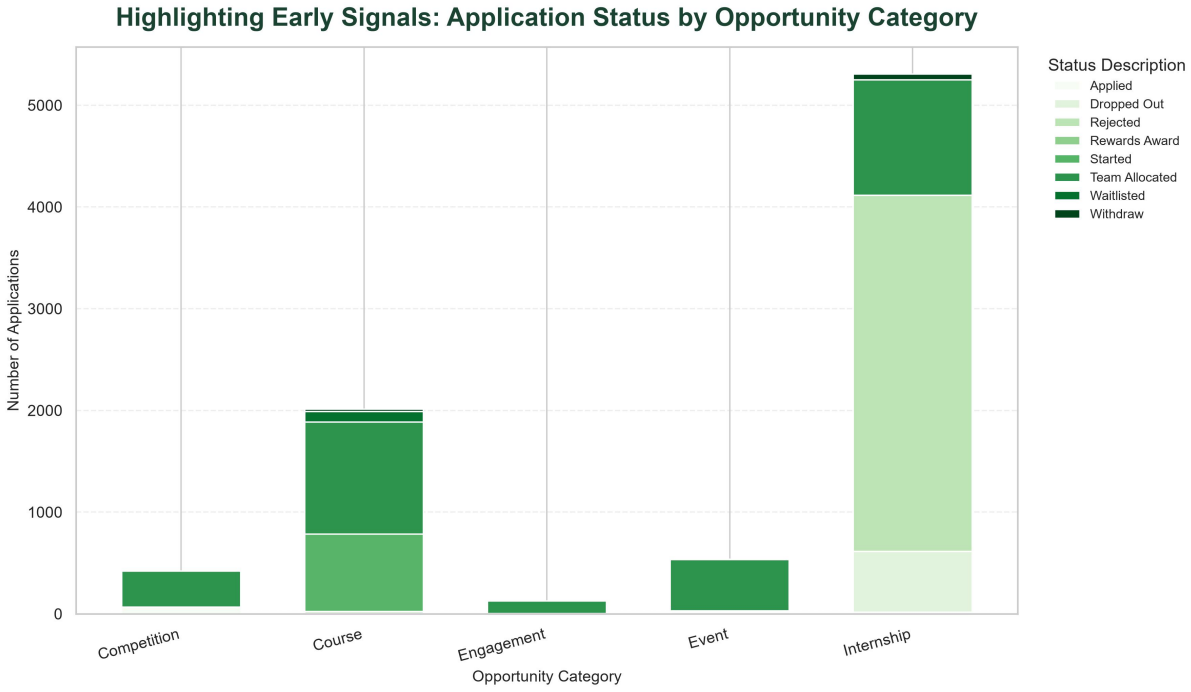
The empirical findings reveal a striking contrast in conversion of velocity across program types. Course applications demonstrate the tightest engagement window, with the vast majority of learners applying within zero to thirty days of post-signup. Conversely, the Internship category exhibits a significantly wider distribution of spread, extending up to over four hundred days. This indicates that while internship offerings generate strong long-term consideration, they require a substantially longer decision-making process. Across all categories, positive outliers exist beyond three hundred days, representing re-engaged users returning to the platform after extended inactive periods.

Strategic Recommendations

To capitalize on early intent, automated nudge campaigns via email or SMS should be deployed within the first fourteen days post-signup for users viewing course offerings. For internship seekers, marketing and operations teams should establish mid-funnel informational touchpoints to address applicant hesitations, provide clarity on requirements and accelerate the overall decision-making timeline.

Application Status by Opportunity Category

This visual evaluates application volume across different stages of the platform lifecycle. By grouping data according to opportunity category and status description, the analysis details how candidates progress through the conversion pipeline. A stacked bar chart visualization was utilized to examine both total contribution volume and the relative proportion of lifecycle stages per category.



Key Insights & Findings

The analysis demonstrates an overwhelming volume concentration within a single program type: the Internship category accounts for over seventy percent of total platform application volume, single-handedly outperforming all other categories combined. However, a significant conversion bottleneck is evident. Despite massive top-of-funnel interest, only a small fraction of applicants successfully transitions from the initial "Applied" stage to terminal positive outcomes such as "Started" or "Rewards Award." Course offerings represent the second-largest volume driver, whereas Event and Engagement categories maintain a minimal overall footprint.

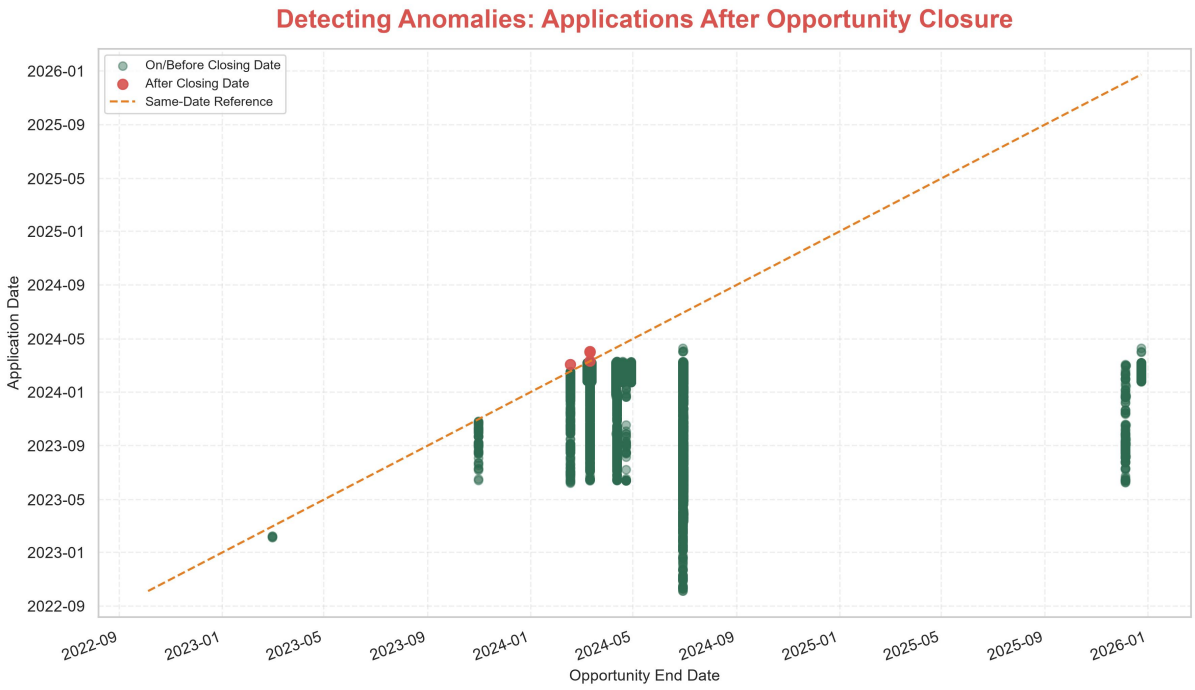
Strategic Recommendations

Cross-functional teams should conduct an immediate operational audit of the internship vetting and review pipeline to identify structural bottlenecks causing candidate drop-off between submission and onboarding. Furthermore, system resources and platform infrastructure should be reallocated to support heavy internship throughput while evaluating whether low-volume categories require strategic repositioning or program sunsetting.

Detecting Anomalies

Applications After Opportunity Closure (Timing Errors)

This visual assesses temporal data integrity by evaluating the alignment between application submission dates and official opportunity closing dates. A strict validation rule was applied to flag any instance where an application date occurred after the published closing date. The relationship was visualized using a dual-variable scatter plot set against a y = x same-date baseline reference line.



Key Insights & Findings

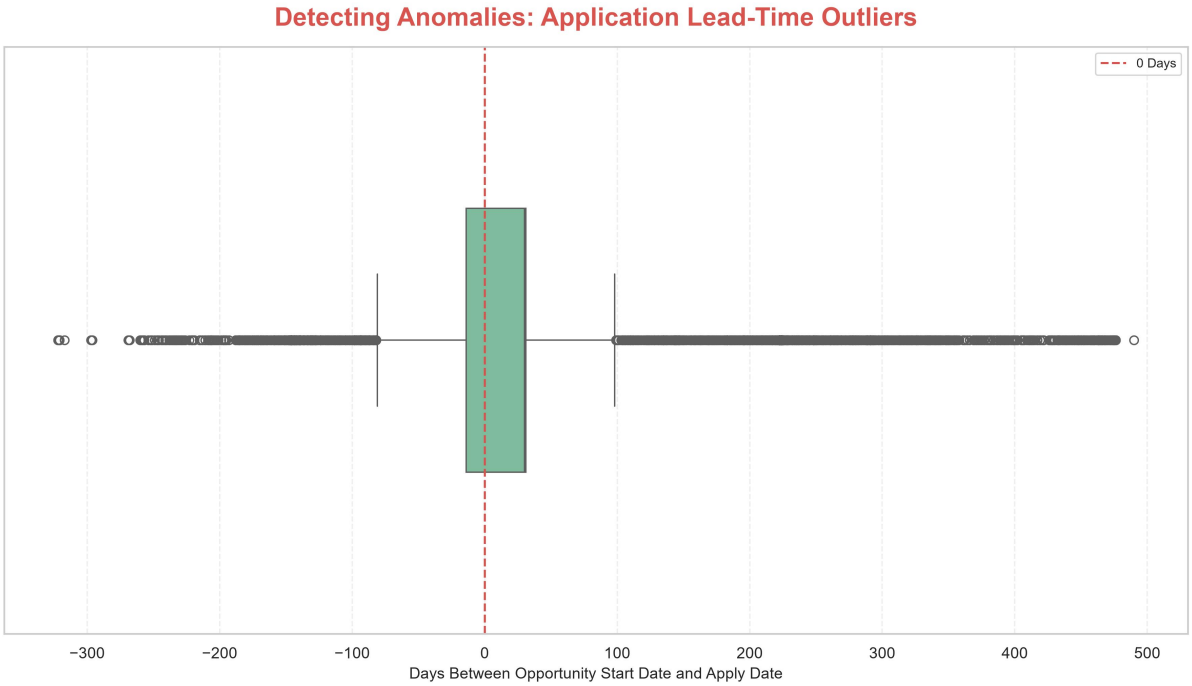
The audit flagged exactly six application records submitted after official opportunity end dates. On the scatter plot visual, these six database records render as three distinct marker clusters due to identical submission timestamps overlapping directly on top of one another. Aside from these specific exceptions, the vast majority of data points sit safely on or beneath the diagonal baseline indicating general compliance with system cutoff dates. While the overall volume of post-close submissions remains low, these records represent critical compliance and system integrity risks

Strategic Recommendations

To mitigate future timing discrepancies, software engineering teams must enforce strict server-side validation rules that automatically lock application forms at 23:59 on the scheduled closing date. Additionally, backend engineering should audit server and database timestamp protocols to ensure universal time zone synchronization (such as UTC enforcement) across all regional operations and batch updates

Application Lead-Time Outliers

This visual examines extreme system anomalies by measuring the interval between opportunity start dates and application dates. The evaluation focused on identifying corrupt negative values where applications were logged prior to opportunity launch dates, as well as long-tail lead-time outliers exceeding one hundred eighty days. A horizontal box plot with individual outlier marker rendering was deployed to isolate these data integrity failures.



Key Insights & Findings

The analysis identified ninety-nine records containing corrupt negative lead times, pointing to database sequence failures, retroactively generated records, or timestamp overwrites. When combined with post-closure submissions, a total of one hundred five data anomalies were identified within the cleaned_data1-feature.xlsx dataset. Furthermore, extreme positive outliers extend beyond five hundred days, which severely distorts standard performance averages if left uncorrected during routine executive reporting.

Strategic Recommendations

Data engineering teams should immediately integrate automated ETL cleaning scripts to filter out negative deltas and cap upper-bound outliers prior to feeding operational dashboards. In parallel, system administrators should conduct an internal log audit on backend record-creation routines to determine why application timestamps are permitted to precede opportunity creation dates.

Quarterly Analysis on Applications

Monthly signup records were then grouped by month to identify changes in participation over time. The monthly results showed variation rather than a consistent upward or downward trend. The highest monthly count occurred in January 2024 with 1,547 records, while August 2023 also showed high activity with 1,133 records**.**
Because monthly figures can fluctuate significantly, the records were also grouped into quarters:

Quarter	Records
2023 Q1	500

2023 Q2	1,435
2023 Q3	2,503
2023 Q4	1,270
2024 Q1	2,739
2024 Q2	11*

*Q2 2024 is incomplete because the dataset only goes into early April 2024 and therefore should not be interpreted as a full-quarter decline.

The quarterly results indicate that participation increased greatly from Q1 to Q3 2023, declined in Q4 2023, and increased again in Q1 2024. Overall, the dataset shows fluctuating participation pattern rather than a consistent long-term growth or decline pattern**.**

Growth and Decline Patterns

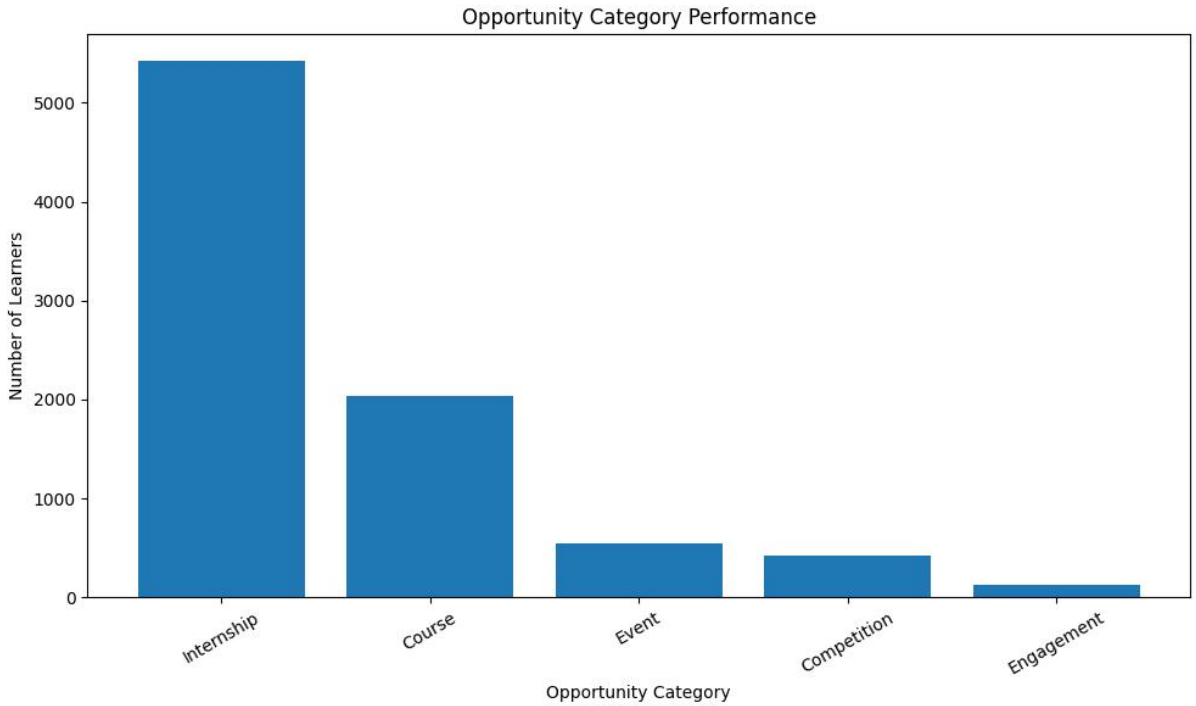
Month-over-month percentage change was calculated to identify periods of particularly strong increases or decreases.
The analysis showed several substantial changes, including:

- A strong increase from March to June 2023.
- A major increase in August 2023.
- Consecutive monthly declines from September through November 2023.
- A strong rebound in December 2023 and January 2024.
- A decline after January 2024.

The large decline observed in March and April 2024 should be interpreted with caution because the dataset may not contain a complete period for these months, particularly April.

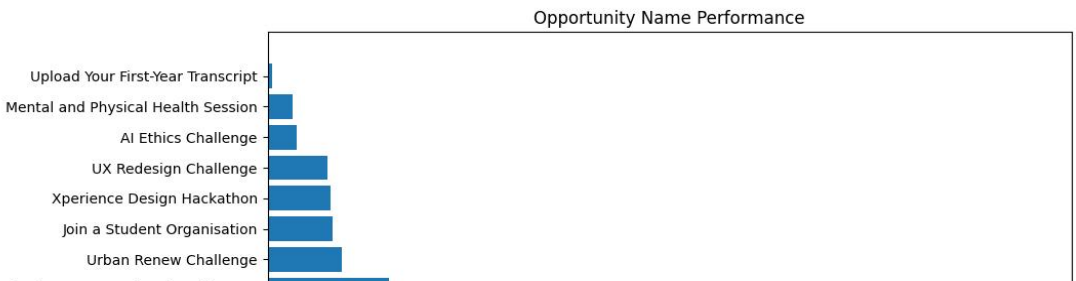
Opportunity Category Analysis

1. Category Performance



The chart shows that internships had the highest number of learners, with 5419 records followed by courses with 2034 learners. Events recorded 545 learners, competition and engagements had 425 and 130 learners. This indicates that internships were the most frequent selected opportunity category, suggesting a stronger level of learner participation in internship opportunities. Engagement had the lowest participation, indicating comparatively lower involvement.

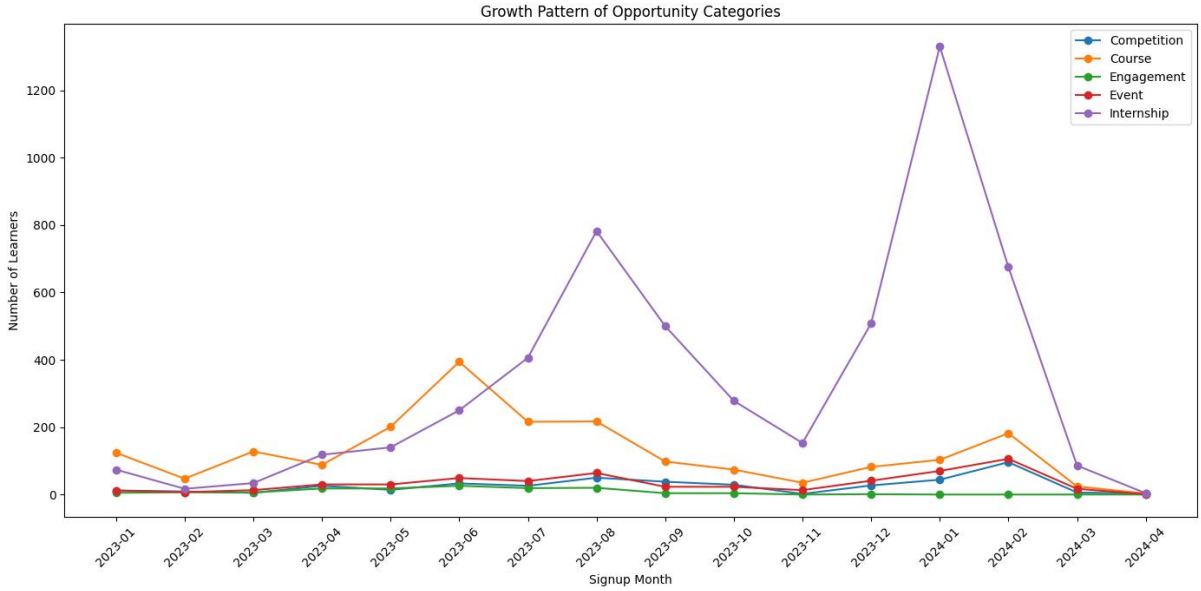
Opportunity Performance



The chart shows that learner participation varies across different opportunity names. Career Essentials had the highest participation with 1421 learners, followed by Data Visualization with 979 learners and Project Management with 836 learners. This indicates strong participation focuses on professional development, technical skills and project management,

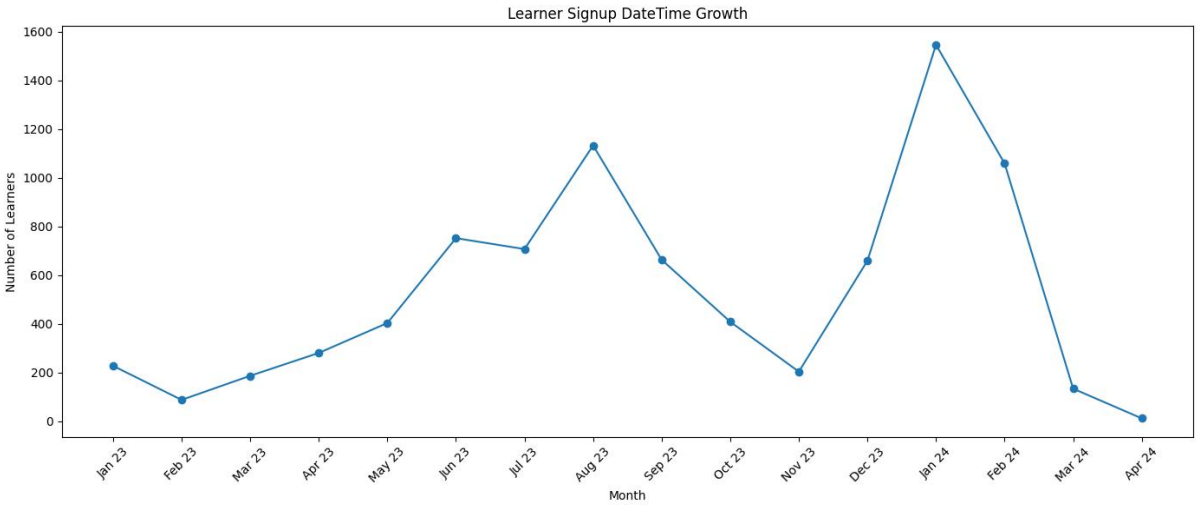
Opportunities such as AI Ethics Challenges (55 learners), Mental and Physical Health Session (46 learners) and Upload Your First-Year Transcript (9 leaners) had the lowest participation.

2. Growth Patterns



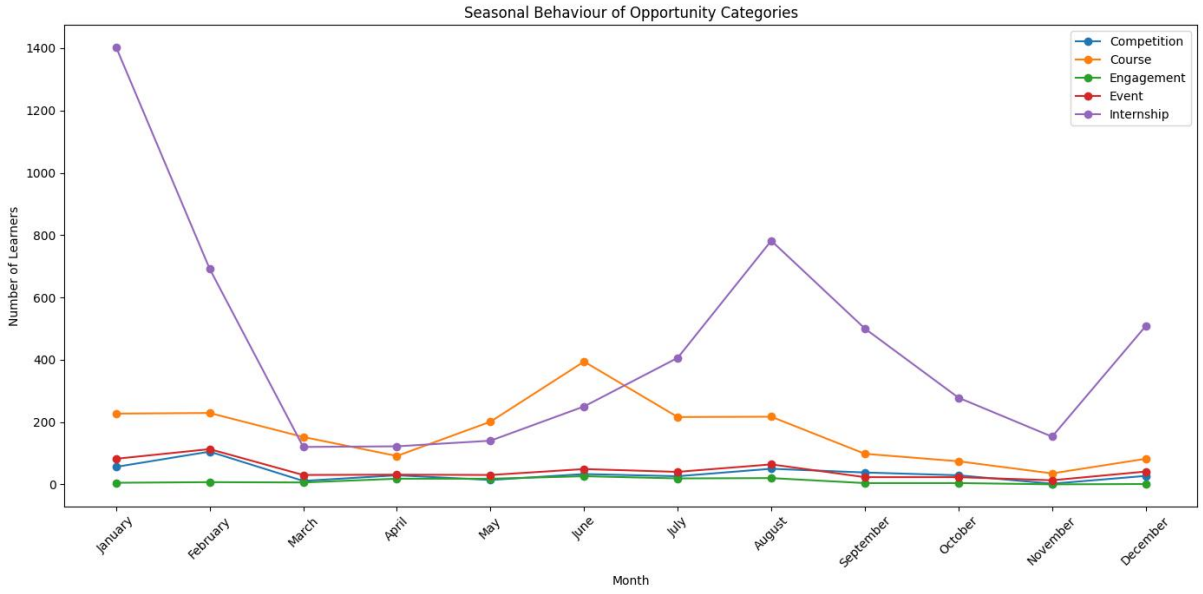
The line chart shows that learner participation was not steady. Internship participation experienced the most growth, increasing from 74 learners in January 2023 to 1330 learners in January 2024. Internships also increased during August 2023, reaching 782 learners. Course participation was highest in June 2023, with 394 learners, while events and competitions showed smaller increases during months. This suggests that learner demand for opportunities varied throughout the period, with internships showing the strongest overall growth pattern.

Learner Signup Date Time Monthly Growth



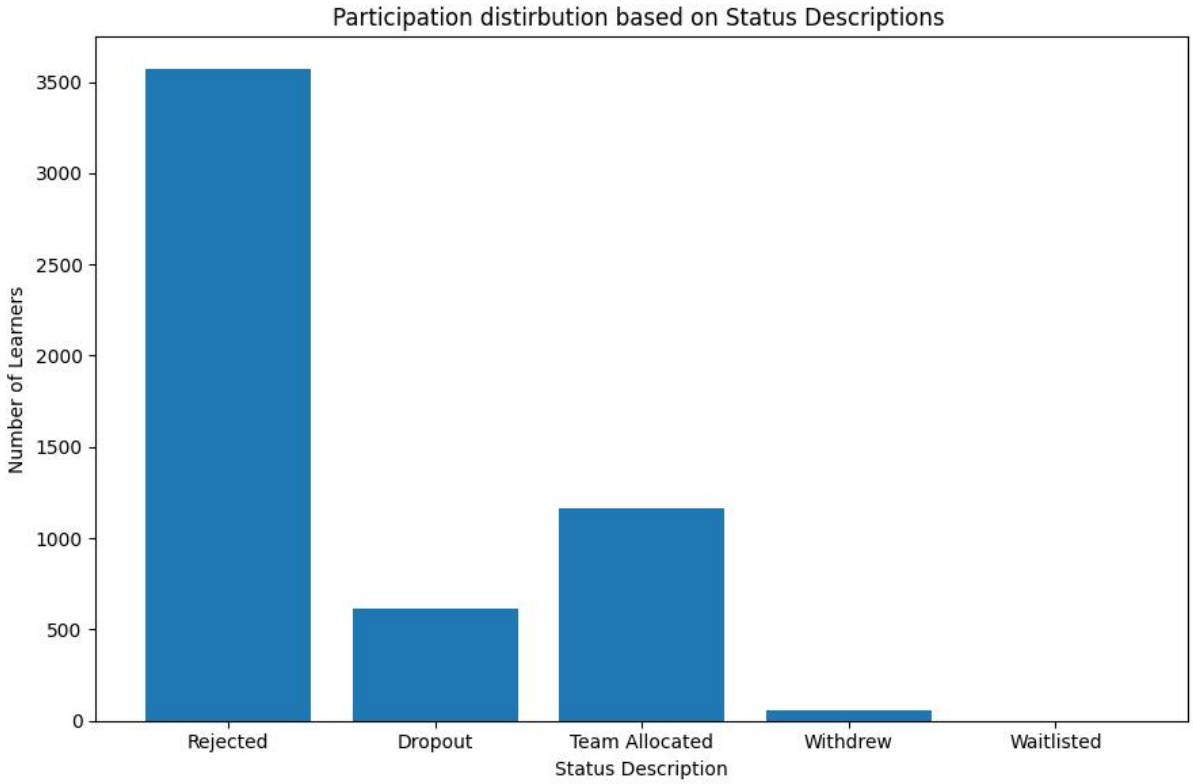
The chart shows that learner signups increased overall during 2023, with a noticeable peak in August at 1133 learners. Signups reached the highest point in January 2024, with 1547 learners, before dropping sharply in March and April 2024 as also discussed in Quarterly Analysis section.

3. Seasonal Behavior



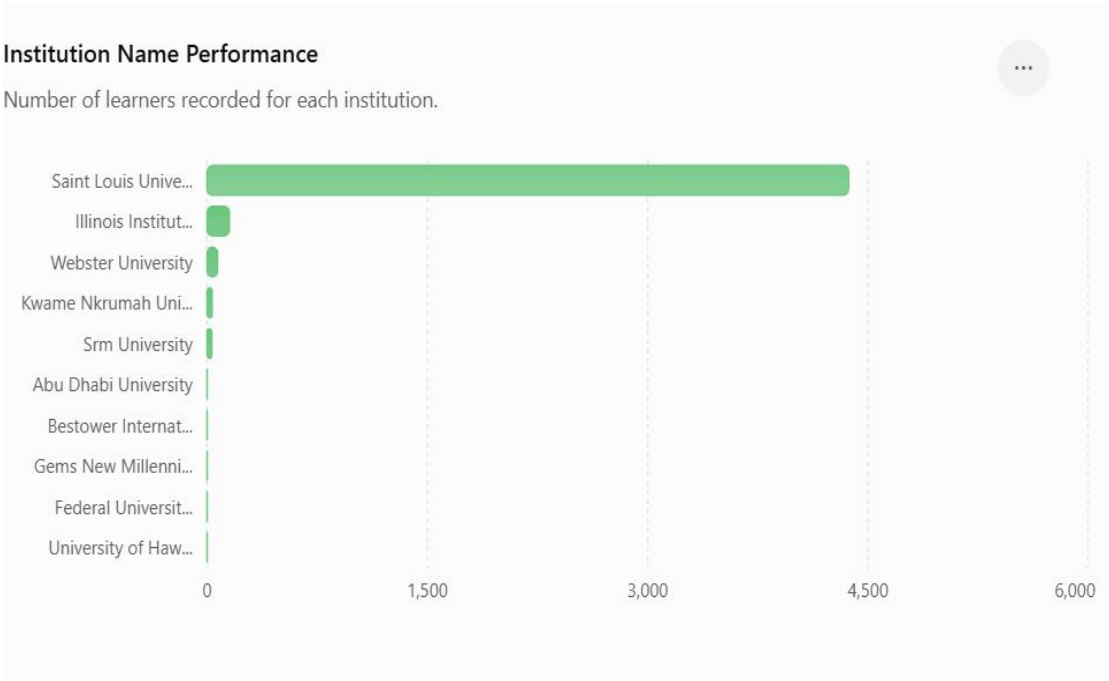
The seasonal chart shows that participation varies depending on the month. Internships displayed the strongest seasonal pattern, reaching a peak of 1404 learners in January, followed by 782 learners in August. Courses had their highest participation in June, with 394 learners, while events and competitions recorded their highest participation in February, with 113 and 105 learners. Engagement remained comparatively low throughout the year. These patterns suggest that learner participation is influenced by periods of the year, with January and August being especially important periods for internship applications.

4. Participation Distribution



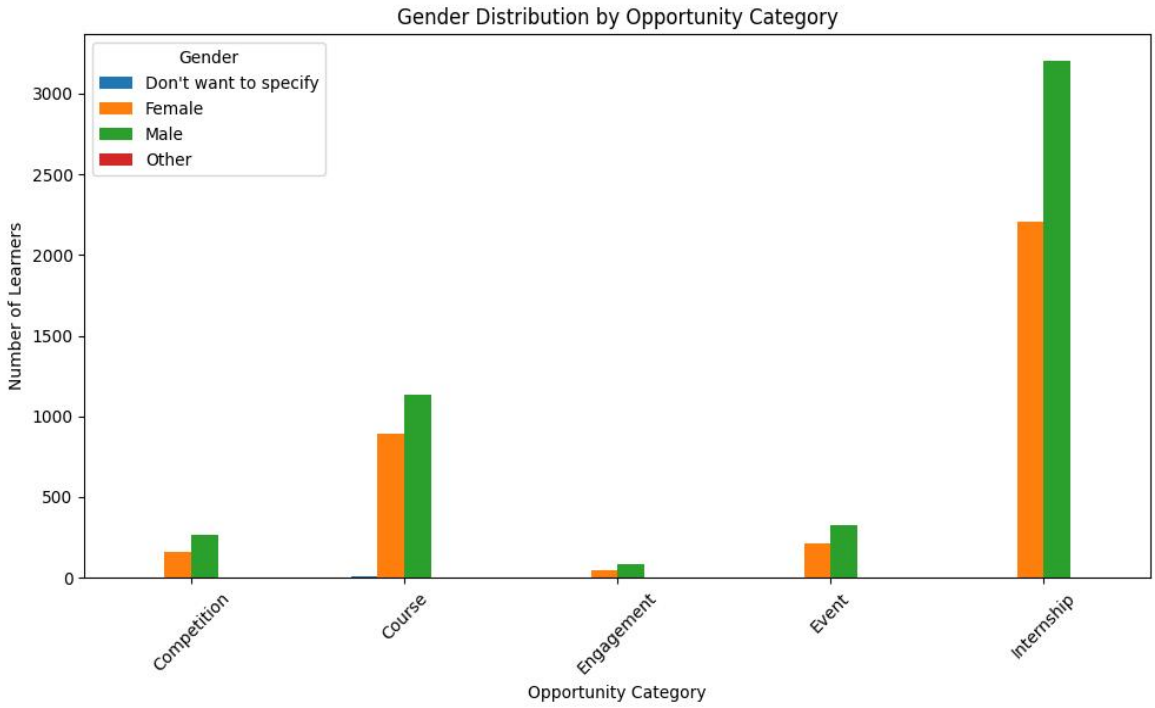
The chart shows different in learner outcomes across opportunity categories. Internships recorded with the largest number of rejects learners, at 3568, as well as 612 learners who dropped out and 1165 who were allocated to a team. Courses had the highest number of learners who started at 764, while events had 513 learners who were team allocated. Competitions recorded 358 team allocations. This suggests that internships had both the highest level of participation and the greatest number of rejections, while courses, events and competitions showed levels of active participation or team allocation

5. Institution Participation



The chart shows a difference in learner participation across institutions. Saint Louis University recorded 4368 learners, which was higher than Illinois Institute of Technology with 152 learners and Webster University with 72 learners. The remaining institutions recorded the same number of learners, which was one. This indicated that Saint Louis University concentrated heavily on participation, while participation from the other institutions was comparatively limited.

6. Gender Distribution by Opportunity Category



The chart shows male learners had the higher participation than female learners across all five opportunity categories. The largest difference occurred in internships, where there were 3207 male learners compared with 2208 female learners. Courses also showed a notable difference, with 1131 male learners compared with 891 female learners. Competition, engagement, and event participation followed the same pattern of higher male participation than female participation. This visual indicates consistent gender participation across the opportunity categories.

Overall Analysis

The visuals show that learner participation was concentrated in internships and were strongly influenced by institution, gender, time of year and learner status. Internships had the largest participation overall and strongest growth and seasonal variation but recorded substantial numbers of rejections and dropouts. Courses, events, and competitions showed different participation and outcome patterns demonstrating learner engagement varied depending on the type of opportunity.

FORECASTING SIGNALS

1. Key Predictor Variables (Model Features)

To forecast learner enrollment, course completion, or application conversion rates accurately, Your model should leverage features across three core tiers:

A. Temporal Lag & Duration Features (Calculated Intervals)

● Application Conversion Lag: Apply Date - Learner Signup Date Time represents the decision latency between a user joining the platform and applying for an opportunity.

● Onboarding Lead Time: {Opportunity Start Date - Apply Date} measures the time buffer before program kick-off, influencing drop-off/withdrawal behaviors.

● Learner Tenure / Program Duration: Opportunity End Date} - {Opportunity Start Date} defines total engagement length.

B. Categorical & Demographic Indicators

● Geography & Academic Profile: Country, Institution Name, and Current/Intended Major (e.g., Computer Science, Information Systems, Healthcare). Regional academic calendars directly drive active windows.

● Demographics: Gender and age derived from Date of Birth relative to application dates.

● Opportunity Attributes: Opportunity Category (e.g., Course, Internship) and Opportunity Name.

C. Target / Status Classifiers

● Learner Status (Status Description / Status Code): Distinct lifecycle stages including Started (Code 1080), Team Allocated (Code 1070), Waitlisted (Code 1040), and Withdraw (Code 1110).

2. Strong Seasonal Indicators & Drivers

Seasonal drivers extracted from quarter features (Quarter Application Date, Quarter Learner Signup Date, Quarter_Opportunity_Start_Date, Quarter_Opportunity_End_Date) govern the Model & cyclical trends:

A. Academic Term Cycles (Q1 & Q3 Spikes):

- Q1 (Jan–Mar): Higher signup volume driven by new term enrollment and career planning resolutions.
- Q3 (Jul–Sep): Strong intake peaks corresponding with fall term orientation across major feeder regions (e.g., United States, India, Nigeria).

B. Quarterly Mismatch / Lag Patterns:

- As shown in the engineered quarter columns, Quarter Learner Signup Date (e.g., Q1/Q2) often leads Quarter Opportunity Star Date (e.g., Q4) by multiple quarters.
- Capturing these lead-time distributions as probability density functions or lag features are critical to preventing lookahead bias.

C. End-of-Quarter Deadline Effects:

- Application dates show clustering around program start thresholds, producing non-. linear time-series surges before major batch start dates.

3. Key Observations for Future Modeling

Conversion Funnel Modeling:
Rather than predicting raw counts directly, implement a multi-stage architecture:
Demand Model: Forecasts daily/weekly signup volume using time-series baseline. methods (ARIMA, Prophet, or temporal XGBoost) incorporating quarterly. seasonality.

Propensity Score Model: Standard binary or multi-class classifier predicting. whether a signed-up learner converts to 'Started' or 'Withdraw' given them demographics, major, and signup-to-apply lag.

B. Data Leakage Mitigation:
Features like Entry created at or final status variables must not be used as raw predictors for time-of-application forecasting; only historical time-stamped state changes should be included.

C. Sparsity & Cardinality Handling:
High-cardinality categorical (Institution Name, Current/Intended Major) should be target-encoded or grouped into broader domain classifications (e.g., STEM vs. Humanities) to maintain feature stability across
future validation splits.

Usage of AI

ChatGPT was used to assist with structuring and refining sentences in the data analysis interpretation.

Prompt: Rewrite and restructure these sentences to make them concise, clear, and suitable for a data analysis report, while maintaining the original meaning and using appropriate analytical terminology.

Prompt: How can I use pandas to calculate and compare the performance of each opportunity category?

Prompt: How can I use pandas to identify seasonal behaviour in learner signups across the different opportunity categories?

Prompt: How can I use pandas to calculate and identify the growth pattern of learner participation across the opportunity categories over time?

Prompt: How can I use pandas to calculate the participation distribution across opportunity categories and status descriptions?

Prompt: Which visualization would be suitable for the following headings: opportunity category performance, growth pattern, seasonal behavior and participation by using the data I have calculated.

Prompt: What pandas/matplotlib should I use to create visualizations from my dataset results?

Google Gemini was used to assist with the analysis of the forecasting signals across sheets including the cleaned dataset (without missing values) and the new features.

Prompt Used:

“Act as a data analyst, based on the structured dataset provided in the spreadsheet, identify. key variables likely to influence predictions, strong seasonal indicators/drivers needed to build accurate, high-confidence future forecasting models, summarize the key observation for future modelling, use credible evidence, do not invent statistics or sources.”